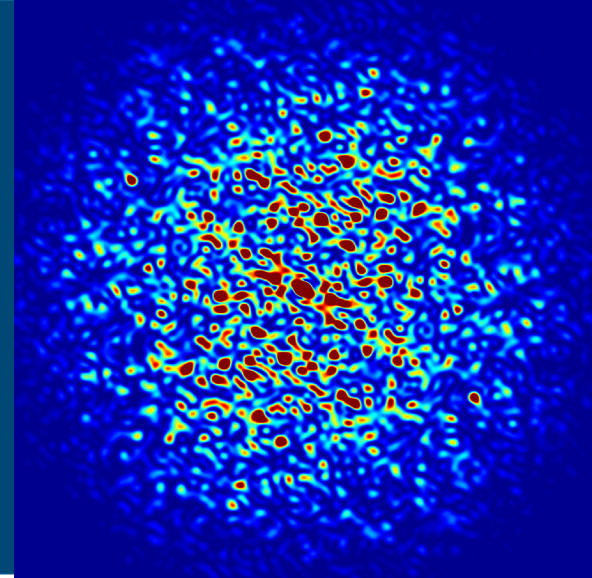


Optimized Linear and Second Order Chromaticity Setpoints for the Advanced Photon Source Upgrade



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2019 North American Particle Accelerator Conference

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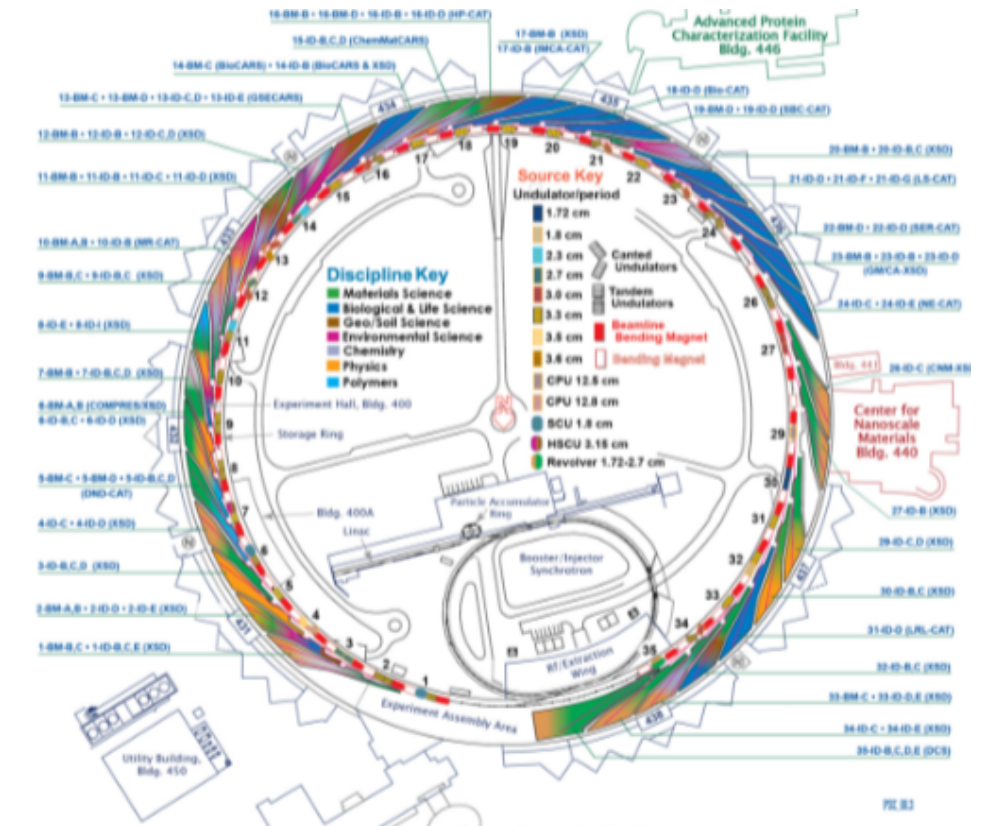
Acknowledgment

- APS-U MAC committee for recommendations on this work
- The authors thank Michael Borland for developing new feature in ELEGANT code of adding quadrupole and skew quadrupole errors in sextupole magnets
- Vadim Sajaev for providing commissioning simulations result
- Michael Borland, Louis Emery, Ryan Lindberg, Vadim Sajaev, Aimin Xiao, C.Y. Yao and other AOP group members for discussions
- Weiming Guo (BNL) for helpful discussions on a similar optimization approach
- Simulations used PELEGANT on weed cluster at APS
- Work supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences, under Contract No. DE-AC02-06CH11357.

Advanced Photon Source (APS)



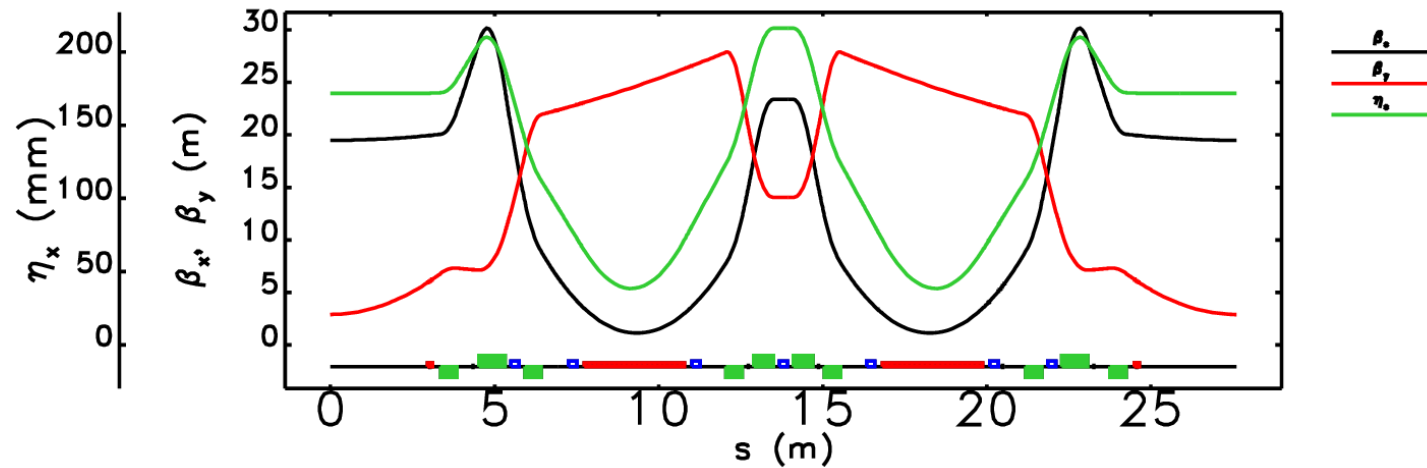
Lemont, IL
Commissioned 1995, 7GeV, 102mA
3nm emittance, 1104m



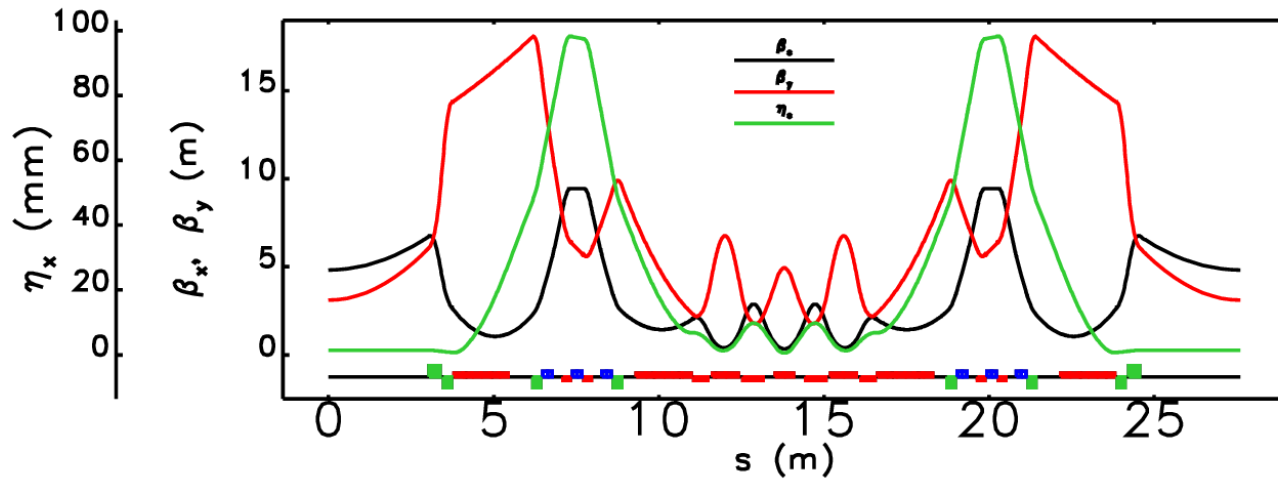
~68 beamlines
More than ~5000 users

Figures from: <https://www.aps.anl.gov/>

Advanced Photon Source Upgrade (APS-U)



APS, 7GeV
DBA cell, 3nm emittance
Nat. Chrom.: **-90/-43**
Max K2*L: **8**

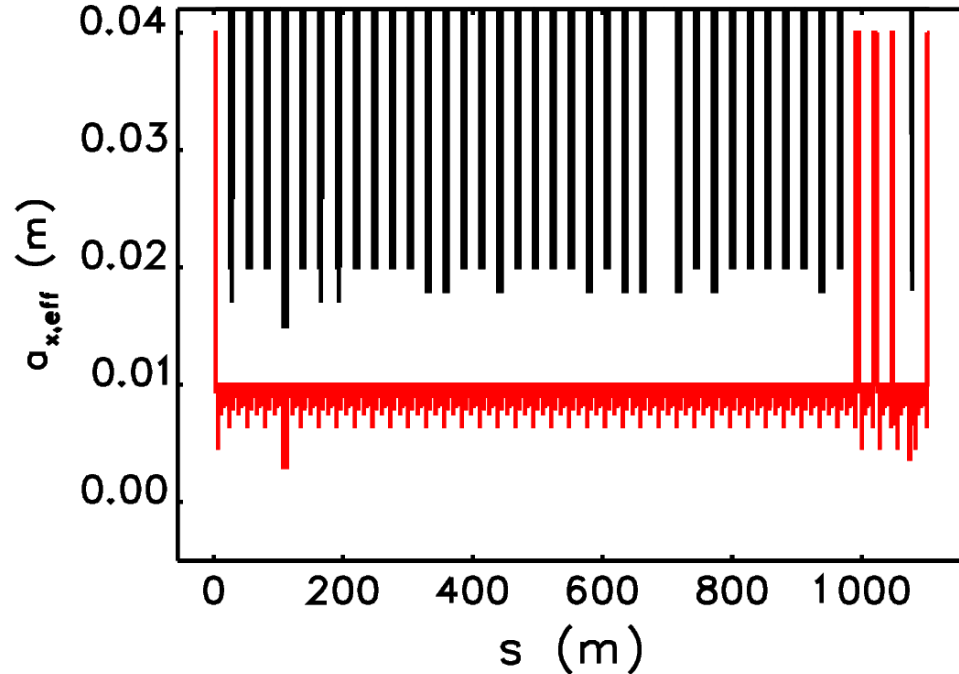


APS-U, 6GeV
HMBA* cell, 42pm emittance
Nat. Chrom.: **-130/-122**
Max K2*L: **66**

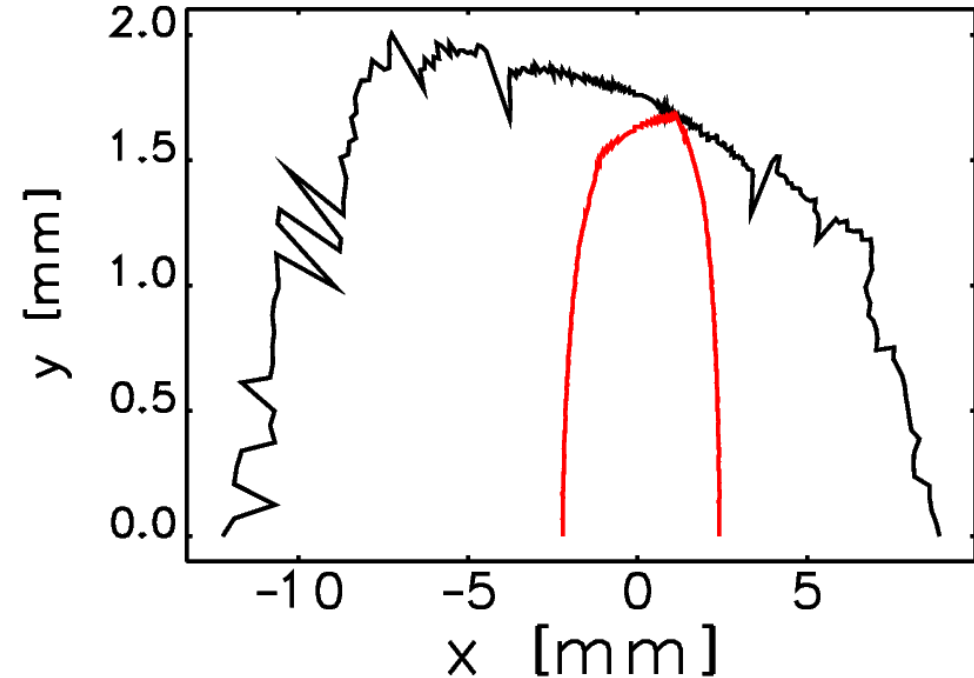
Elements in blue are sextupoles. 7/6 per cell for APS/APS-U

*Concept developed by ESRF

APS-U more challenging than APS

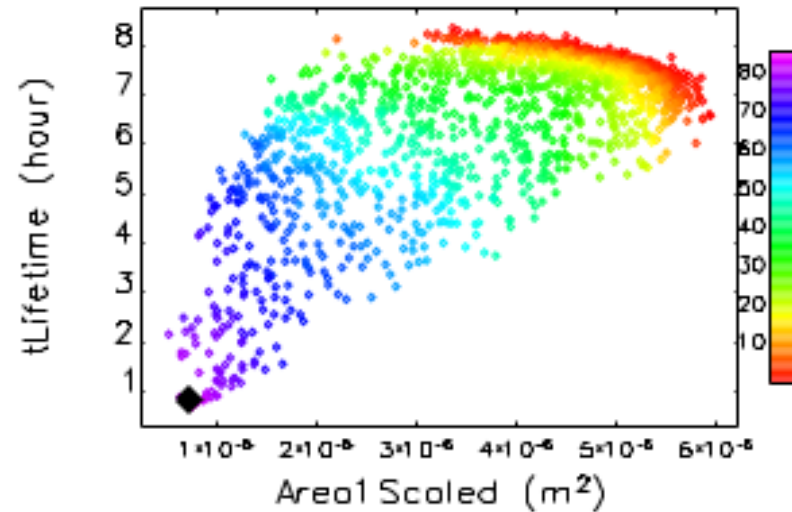


Stronger magnets requires narrow pole gaps → smaller horizontal physical aperture
Red: APS-U; Black: APS



Dynamic acceptance comparison
Red: APS-U; Black: APS

Nonlinear optics optimizations



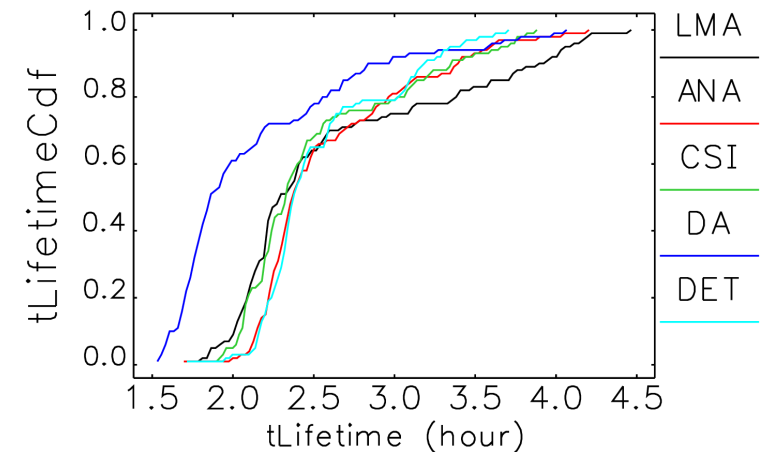
MOGA: LMA, DA, chromatic tunes

M. Borland et al., ANL/APS/LS-319 (2010).

Case	Inj. Eff.	Lifetime [h]
UBOP + K2(UBOP)	87%	69
HSCU + K2(UBOP)	85%	58
HSCU + K2(LMA)	87%	73
HSCU + K2(DET)	92%	66

Verified by APS operations with good agreement on lifetime and injection eff.

Method	Computing time [cores*hour]	Note
ANA	0.04	
CSI	0.23	64 turns + 1 pass
DET	0.71	64 turns
DA	6	50 turns + 3 momentum
LMA	46	400 turns (≥ 1 SO period)



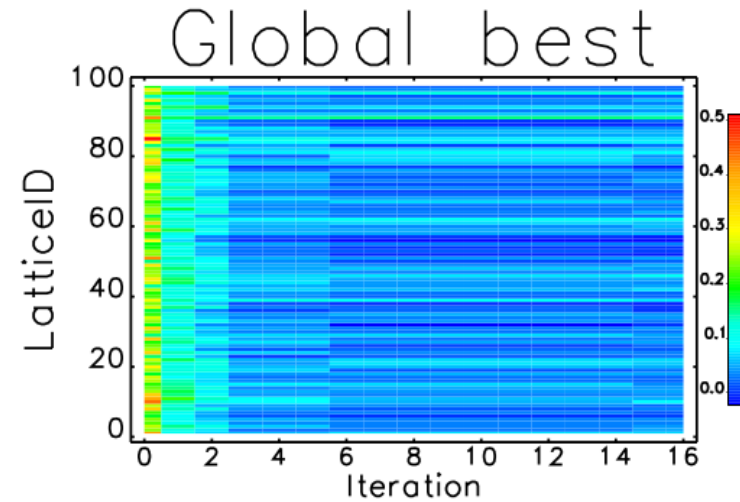
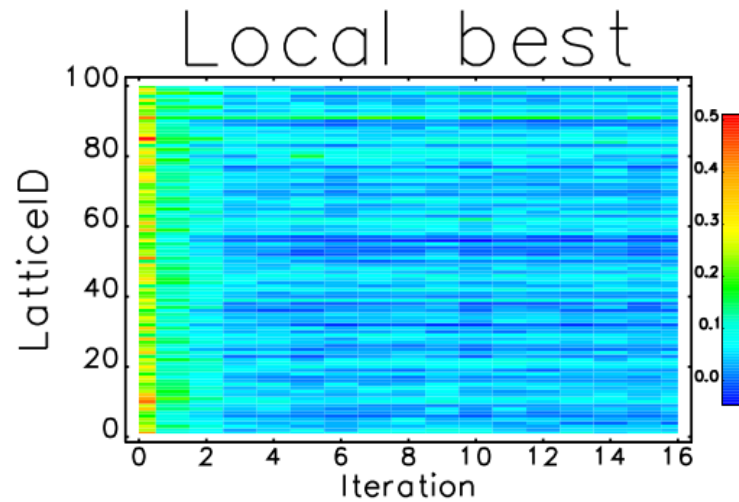
Faster methods developed/benchmarked
Robust, evaluated with diff. objectives, LMA/DA

Y.-P. Sun et al., in NAPAC2016 (2016).

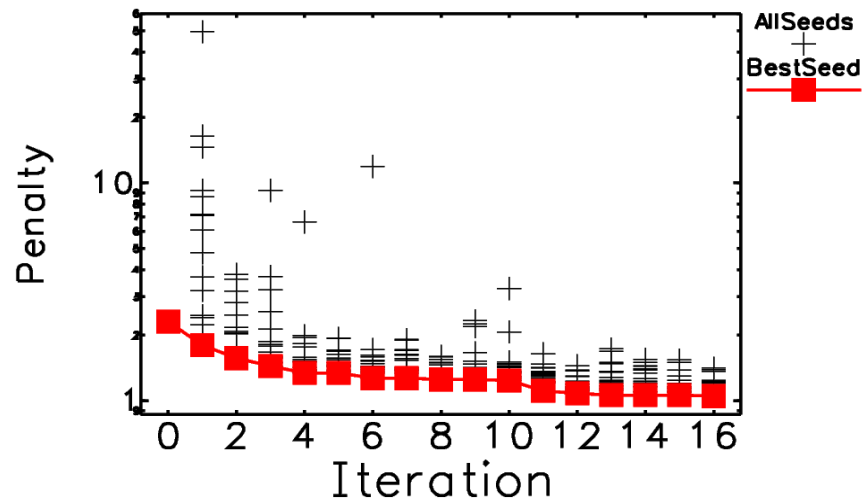
Why and how to change chromaticity

- Chromaticities greatly impacts on single particle dynamics and multi-particle dynamics
 - Important online tuning knobs
 - At APS, linear chromaticity is tweaked to suppress instabilities
- Linear chromaticity knob existed (naturally) for APS-U
 - Can use all sextupoles in 2 families for HMBA
 - Solution not guaranteed to be optimized
 - Nonlinear chrom. knobs harder to formulate
- Here it is proposed to optimize at each chromaticity setpoint, with direct tracking simulations
 - Guaranteed performance
 - Setpoints may not be continuous/smooth in K2

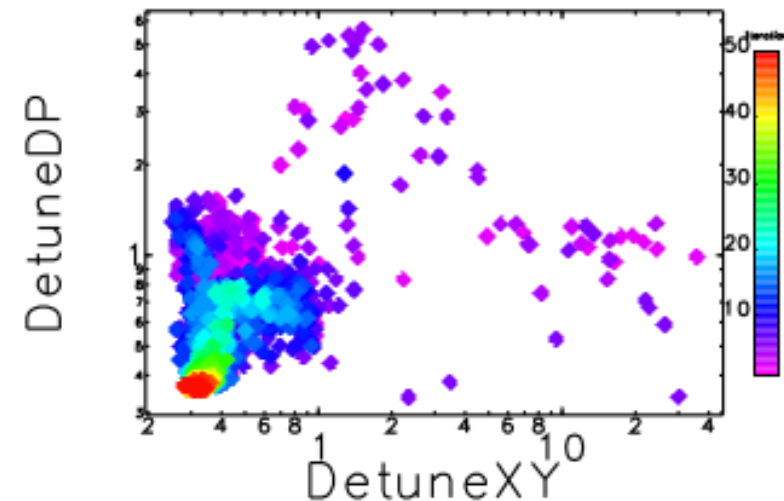
Optimize against many error seeds with DET method



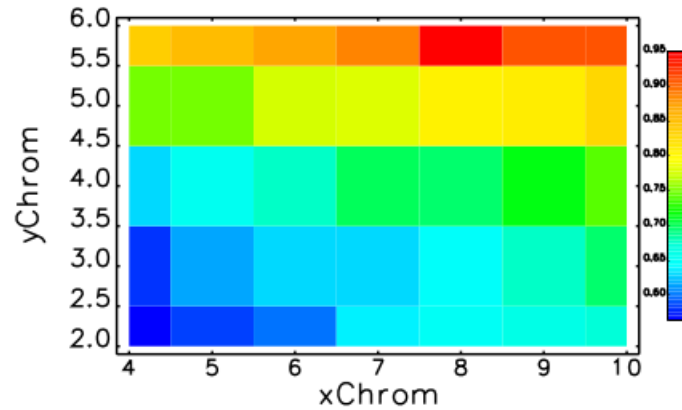
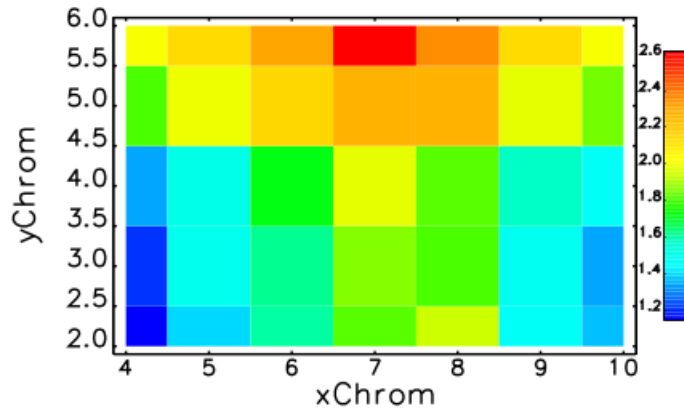
Penalty in iterations



Pareto front

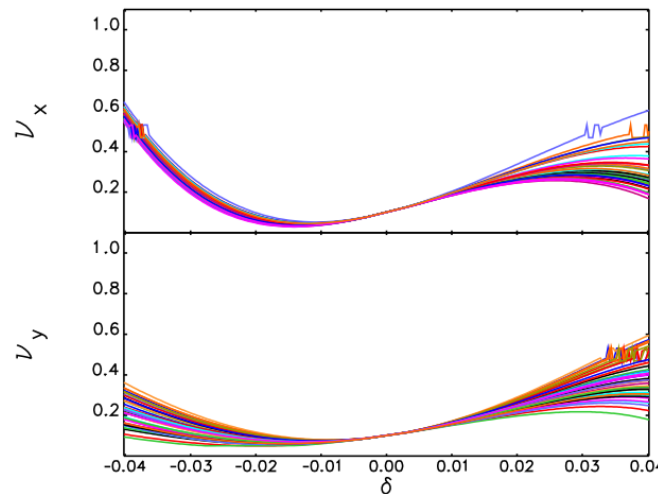
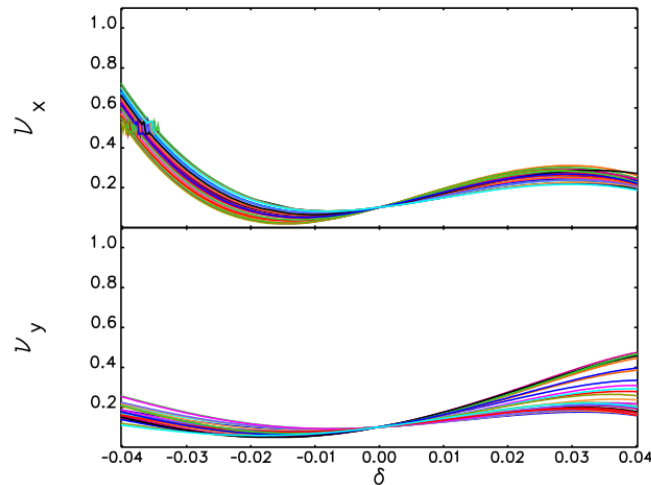


Optimized setpoints for linear/2nd order chromaticity



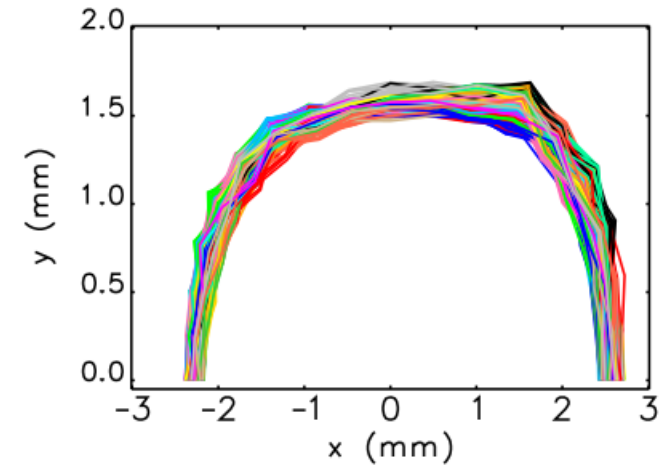
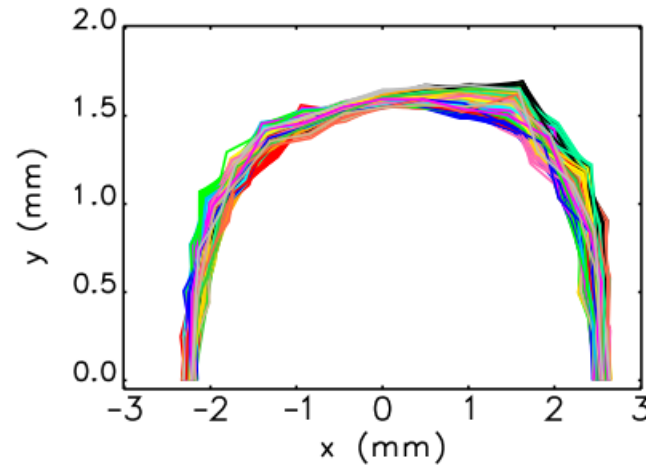
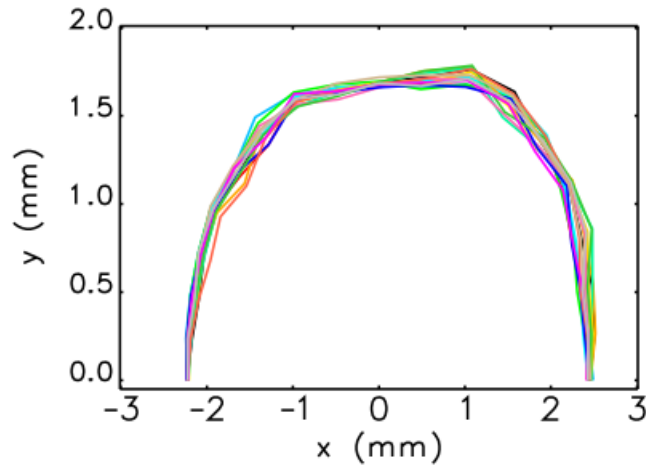
Control K2 range for more smooth
linear chrom. setpoints

Sort/filter for second order

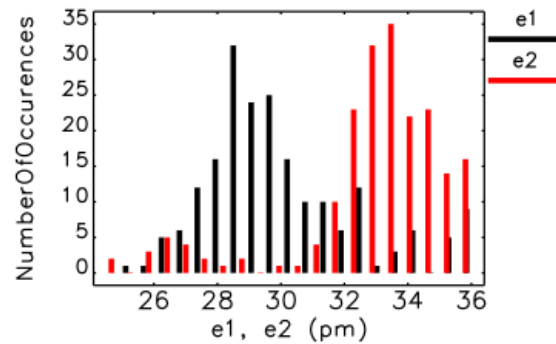
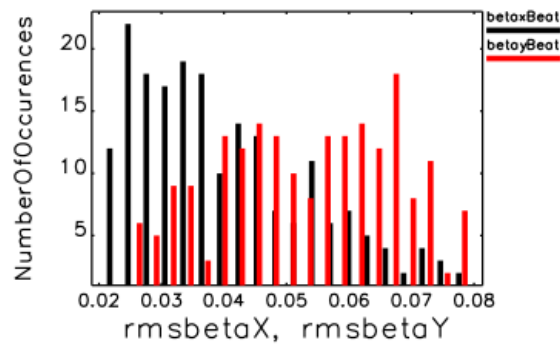


Chromatic detuning for linear and
second order chrom. setpoints

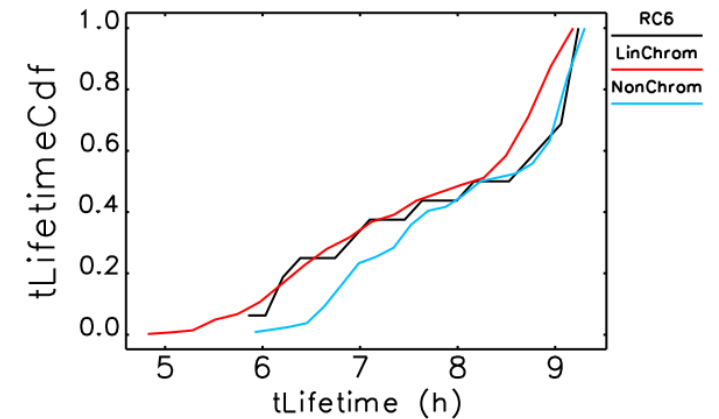
Good performance compared to nominal APS-U



Dynamic acceptance. Left: nominal APS-U; Middle: linear chrom.; Right: second order

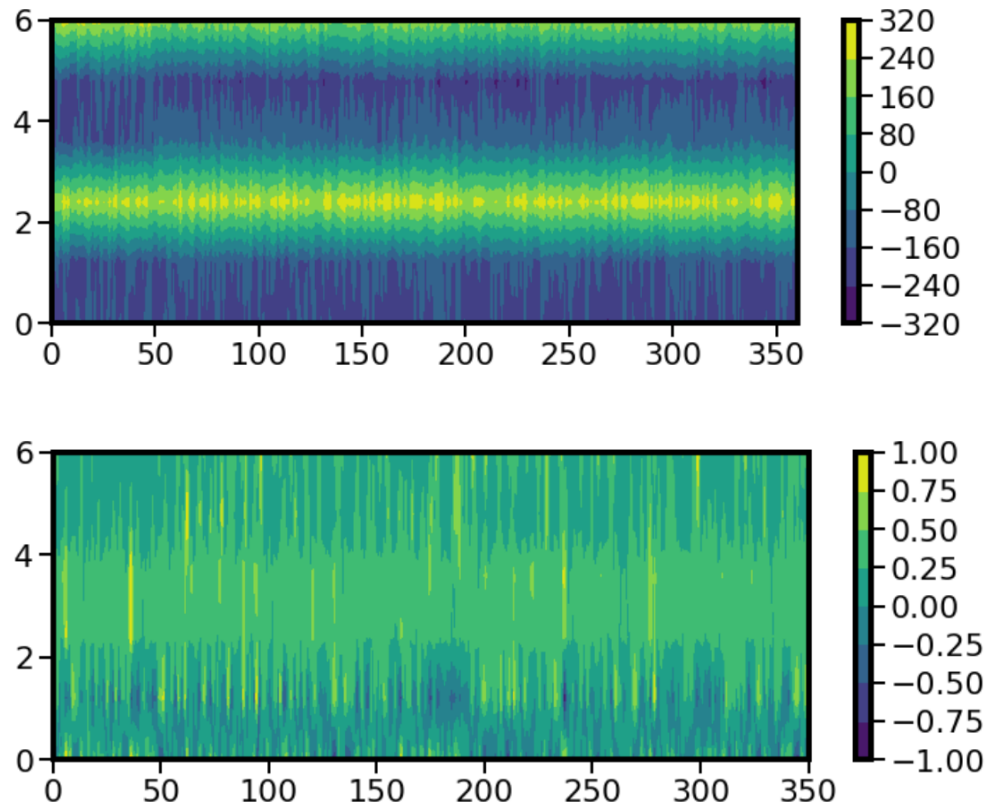


200 error seeds

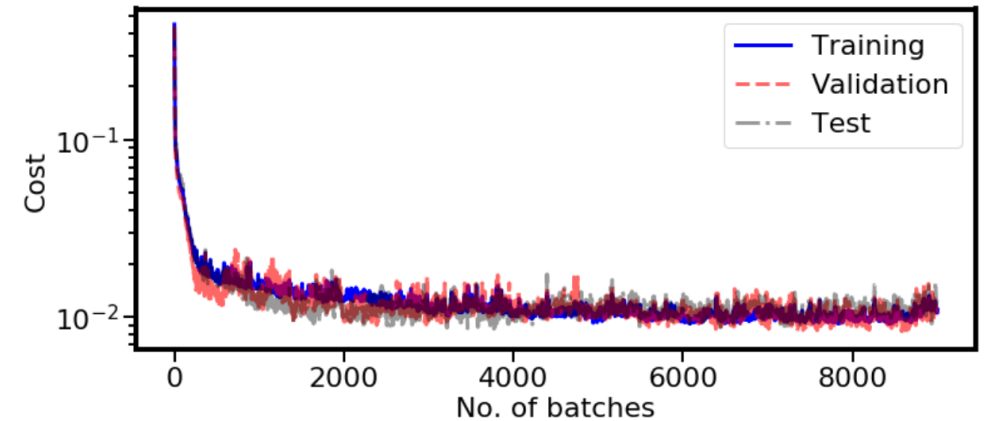
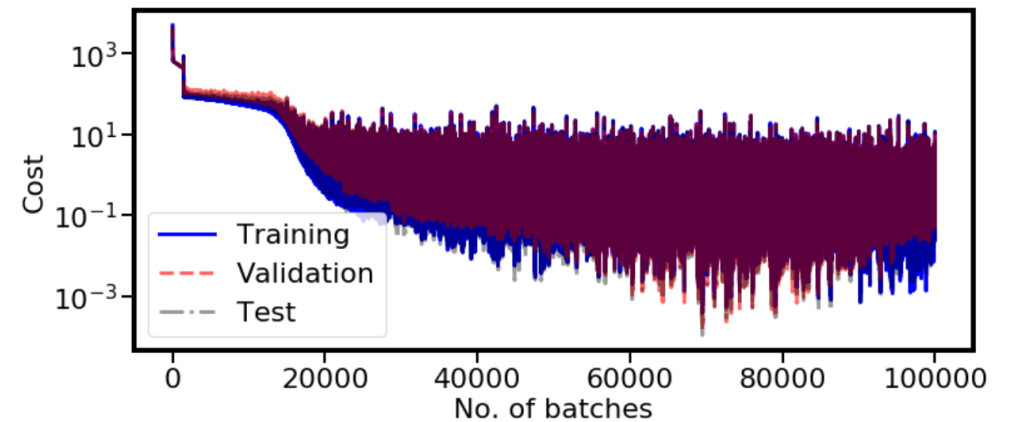


Lifetime

Neural network model



Training data samples
Input (top), output (bottom)



Cost of training/val/test datasets

Discussed one approach to generate optimized linear and second order chromaticity setpoints for the Advanced Photon Source Upgrade

Optimization methods verified by APS operations with good agreement on lifetime and injection efficiency

Thank you for your attention

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